

Lab Resource Utilization Platform

1. Title

Lab Resource Utilization Platform: A Full-Stack Lab Resource Utilization Platform that allows institutions to share expensive laboratory equipment and optimize utilization rates, enabling research institutions, universities, and laboratories to manage equipment inventory, schedule shared resource access, monitor real-time utilization, track maintenance workflows, and analyze resource efficiency through a centralized intelligence dashboard.

2. Objective

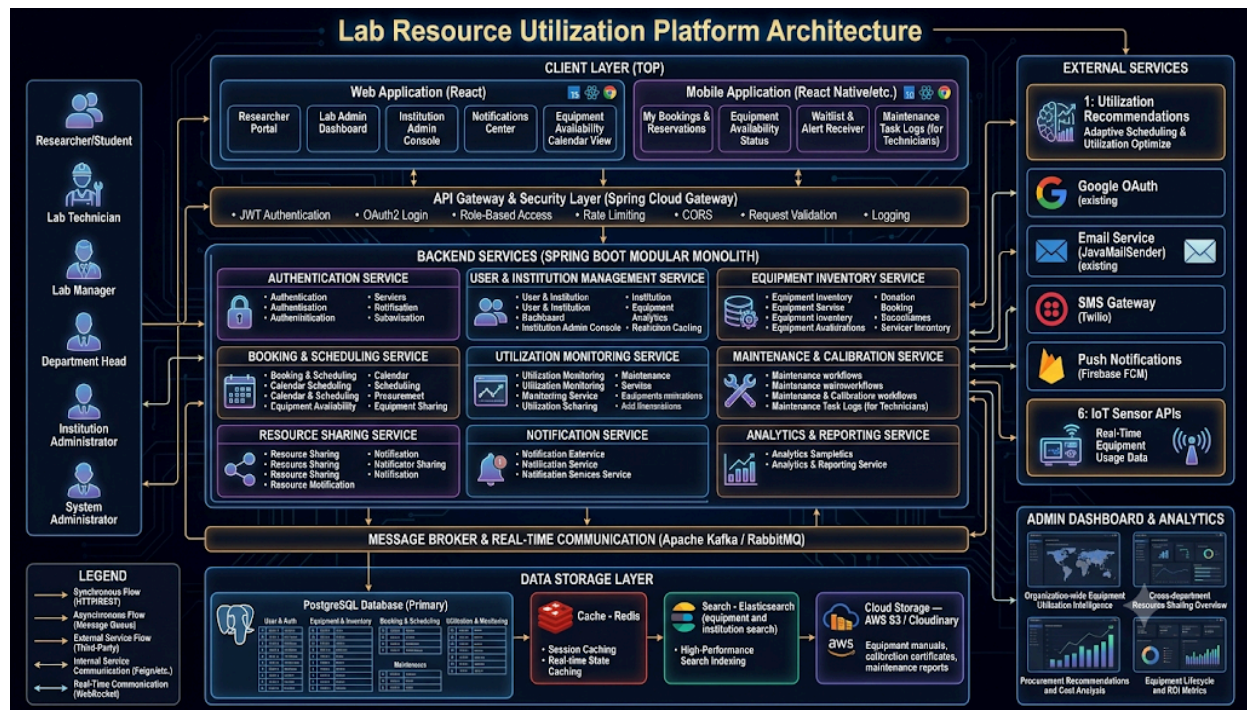
Build a full-stack web application (React.js frontend + Spring Boot backend) that allows research institutions, lab administrators, and faculty members to manage laboratory equipment across departments and institutions, schedule and share access to expensive resources, monitor real-time equipment utilization, handle maintenance and calibration workflows, and drive data-driven decisions on equipment procurement and sharing. The platform provides real-time equipment availability tracking, rule-based scheduling optimization, inter-institution resource sharing workflows, booking and scheduling management, maintenance tracking, and lab analytics through an integrated intelligence dashboard.

Outcomes

- (i) Developed and deployed a full-stack React + Spring Boot application.
- (ii) Implemented secure user authentication using JWT and OAuth2.
- (iii) Built equipment inventory management and resource cataloging workflows.
- (iv) Implemented real-time equipment availability tracking and utilization monitoring.
- (v) Added rule-based scheduling optimization and utilization recommendations.

- (vi) Created inter-institution resource sharing and booking workflows.
- (vii) Implemented analytics dashboards and equipment efficiency reporting modules.
- (viii) Deployed frontend and backend using PostgreSQL in production.
- (ix) Added notification systems for booking confirmations, maintenance alerts, and utilization updates.

3. Architecture Components



Client Layer (React.js)

- Web Application
- Researcher Portal
- Lab Admin Dashboard
- Institution Admin Console
- Notifications Center
- Equipment Availability Calendar View

API Gateway & Security Layer

- JWT Authentication · OAuth2 Login · Role-Based Access · Rate Limiting · CORS · Request Validation · Logging

Backend Services (Spring Boot Modular Monolith)

- Authentication Service
- User & Institution Management Service
- Equipment Inventory Service
- Booking & Scheduling Service
- Utilization Monitoring Service
- Maintenance & Calibration Service
- Resource Sharing Service
- Notification Service
- Analytics & Reporting Service

External Services

- Google OAuth
- Email Service (JavaMailSender)
- SMS Gateway (Twilio)
- Push Notifications (Firebase FCM)
- IoT Sensor APIs (real-time equipment usage data)

Cache & Messaging

Cache — Redis

Message Broker & Real-Time Communication — Apache Kafka / RabbitMQ

Data Storage Layer

PostgreSQL Database (Primary):

- User & Auth Tables
- Equipment & Inventory Tables
- Booking & Scheduling Tables
- Utilization & Monitoring Tables
- Maintenance Tables
- Notification Tables

- Analytics Tables

Cloud Storage – AWS S3 / Cloudinary (equipment manuals, calibration certificates, maintenance reports)

4. Modules to be Implemented

1. User Authentication & Role-Based Access

(i) JWT Authentication (ii) OAuth2 Login (iii) Password Reset (iv) Profile Management (v) Role Management

Roles

- Researcher / Student
- Lab Technician
- Lab Manager
- Department Head
- Institution Administrator
- System Administrator

2. Equipment Inventory Management Module

(i) Equipment registration and cataloging (ii) Equipment specifications and documentation management (iii) Equipment categorization and tagging (iv) Equipment availability status tracking (v) Calibration and certification record management (vi) Department and institution mapping

Equipment Status

- Available
- Booked
- Under Maintenance
- Out of Service
- Retired

3. Booking & Scheduling Module

(i) Real-time equipment availability calendar (ii) Equipment reservation and booking workflows (iii) Recurring booking management (iv) Booking approval and rejection workflows (v) Waitlist management for high-demand equipment (vi) Booking history and audit trail

Booking Status

- Pending Approval
- Confirmed
- In Use
- Completed
- Cancelled
- No Show

4. Utilization Monitoring Module

(i) Real-time equipment usage tracking (ii) Utilization rate calculation per equipment (iii) Department-level and institution-level utilization aggregation (iv) Idle time detection and alerts (v) Peak usage pattern analysis (vi) Utilization heatmap visualization

Utilization Dimensions

- Individual Equipment vs. Capacity
- Department vs. Institutional Targets
- Current Utilization vs. Historical Benchmarks
- Shared vs. Exclusive Usage Patterns

5. Inter-Institution Resource Sharing Module

(i) Cross-institution equipment listing and discovery (ii) Sharing agreement and access request workflows (iii) External booking and access management (iv) Usage fee and cost-sharing calculations (v) Shared equipment scheduling coordination (vi) Sharing analytics and partnership reporting

6. Maintenance & Calibration Module

(i) Preventive maintenance scheduling (ii) Maintenance request and work order management (iii) Calibration tracking and certification renewal reminders (iv) Maintenance history and service logs (v) Equipment downtime tracking (vi) Technician assignment and task management

7. Notification & Alert Module

(i) Booking confirmation and reminder notifications (ii) Equipment availability alerts for waitlisted users (iii) Maintenance due and calibration expiry alerts (iv) Idle equipment

alerts for lab managers (v) Sharing request and approval notifications (vi) Email notifications (vii) SMS notifications (viii) Push notifications

8. Analytics Dashboard Module

Researcher / Student Dashboard

(i) My bookings and upcoming reservations (ii) Equipment availability overview (iii) Booking history and usage summary (iv) Waitlist status and notifications (v) Equipment recommendations based on booking history

Lab Manager / Department Head Dashboard

(i) Department equipment utilization heatmap (ii) Booking adoption and no-show rates (iii) Maintenance schedule overview (iv) High-demand equipment alerts (v) Sharing requests and approvals

Institution Admin Dashboard

(i) Organization-wide equipment utilization intelligence (ii) Cross-department resource sharing overview (iii) Procurement recommendations and cost analysis (iv) Equipment lifecycle and ROI metrics (v) System monitoring and user management (vi) Reports management

9. Cost & Billing Management Module

(i) Usage-based cost tracking per equipment (ii) Department-wise cost allocation (iii) Inter-institution billing for shared equipment (iv) Cost recovery and chargeback workflows (v) Budget utilization tracking (vi) Financial reporting and invoice generation

10. Reports & Export Module

(i) Equipment utilization reports (ii) Department resource usage reports (iii) Maintenance and downtime reports (iv) Inter-institution sharing reports (v) Procurement and cost analysis reports (vi) PDF export (vii) Excel export

11. Final Integration, Testing & Deployment

5. Week-wise Module Implementation and High-Level Requirements

Milestone 1: Week 1 & 2 — Project Initialization, Design Process & Core Setup

Tasks

(i) Define laboratory resource management workflows and project objectives. (ii) Design system architecture and database schema (equipment, bookings, utilization, maintenance). (iii) Create UI wireframes and equipment catalog planning. (iv) Setup React frontend and Spring Boot backend. (v) Implement JWT authentication and role-based access control. (vi) Build equipment inventory management and cataloging module. (vii) Develop booking and scheduling core workflows.

Outcomes

(i) Understand lab resource management and equipment sharing workflows. (ii) Learn system architecture and database design for inventory and scheduling systems. (iii) Build frontend and backend project initialization. (iv) Working authentication and equipment inventory management system.

Milestone 2: Week 3 & 4 — Utilization Monitoring & Inter-Institution Sharing

Tasks

(i) Build real-time equipment utilization tracking and monitoring. (ii) Implement utilization heatmap visualization and idle time detection. (iii) Develop inter-institution resource sharing and access request workflows. (iv) Build external booking and access management for shared resources. (v) Implement utilization rate calculations and demand analysis. (vi) Develop waitlist management and booking optimization logic.

Outcomes

(i) Implement real-time utilization monitoring and visualization. (ii) Build inter-institution resource sharing workflows. (iii) Understand resource allocation and demand analysis concepts. (iv) Generate actionable equipment utilization insights.

Milestone 3: Week 5 & 6 — Maintenance, Cost Management & Analytics

Tasks

(i) Build maintenance scheduling and work order management workflows. (ii) Implement calibration tracking and certification renewal reminders. (iii) Develop cost tracking and inter-institution billing workflows. (iv) Build analytics dashboards for researchers, lab managers, and admins. (v) Generate utilization effectiveness and cost analysis reports. (vi) Implement notification and alert systems.

Outcomes

(i) Build maintenance and calibration management ecosystem. (ii) Implement lab analytics and resource intelligence dashboards. (iii) Understand equipment lifecycle and cost management concepts. (iv) Complete end-to-end lab resource utilization workflows.

Milestone 4: Week 7 & 8 — Testing, Deployment & Documentation

Tasks

(i) Perform application testing and workflow validation. (ii) Improve platform performance and UI responsiveness. (iii) Deploy platform using Docker and cloud environments. (iv) Prepare final project documentation and presentation. (v) Demonstrate the complete Lab Resource Utilization Platform.

Outcomes

(i) Gain deployment and testing experience. (ii) Improve platform stability and usability. (iii) Complete live deployment and final demonstration. (iv) Prepare professional project documentation and presentation.

6. Evaluation Criteria

Milestone 1 (Week 2)

- (i) Project initialization and architecture setup completed. (ii) Authentication and role-based access control implemented. (iii) Equipment inventory management and booking system functional. (iv) System design and UI planning completed.

Milestone 2 (Week 4)

- (i) Real-time utilization monitoring and heatmap visualization working. (ii) Inter-institution resource sharing workflows implemented. (iii) Waitlist management and booking optimization working. (iv) Utilization rate calculations and demand analysis functional.

Milestone 3 (Week 6)

- (i) Maintenance and calibration management workflows implemented. (ii) Cost tracking and inter-institution billing module functional. (iii) Analytics dashboards and utilization reports generated. (iv) Notification and alert system integrated.

Milestone 4 (Week 8)

- (i) Fully deployed frontend and backend. (ii) Testing and validation completed. (iii) Documentation and presentation prepared. (iv) Successful end-to-end platform demonstration completed.

7. Tools & Tech Stack

Programming Language

Backend — Java, Spring Boot

Frontend — JavaScript, React.js

Database

Primary Database — PostgreSQL

Caching — Redis

Search — Elasticsearch (equipment and institution search)

Authentication

Spring Security, JWT Authentication, OAuth2 Login

Libraries & Frameworks

Backend — Spring Boot · Spring Security · Spring Data JPA · Hibernate · Maven · Spring WebSocket

Frontend — React.js · React Router · Axios · Tailwind CSS · Chart.js · Recharts · D3.js · FullCalendar.js · Context API

Notifications

Firebase Cloud Messaging · JavaMailSender · Twilio SMS

Storage

AWS S3 · Cloudinary

Testing

JUnit · Mockito · Postman · React Testing Library

Dev & Deployment Tools

IntelliJ IDEA · VS Code · Git & GitHub · Docker · Docker Compose · GitHub Actions · AWS
/ Azure · Nginx · Postman